

CHARLESTON INTL, SC, USA

WMO: 722080

Lat: 32.8950N

Lon: 80.0275W

Elev: 47

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.2	30.6	8.8	8.6	36.1	13.9	11.2	38.9	25.6	61.9	22.8	58.4	5.8	30	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	94.4	77.8	92.3	77.4	90.3	76.9	80.5	88.8	79.7	87.8	78.7	86.4	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	148.1	84.1	77.5	143.1	83.2	76.7	139.2	82.6	44.2	89.1	43.1	87.5	42.2	86.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	18.3	16.5	20.6	98.2	3.2	2.0	18.4	99.7	16.5	100.9	14.7	102.0	12.4	103.5	
			DB												
			WB	18.1	82.7	3.1	1.1	15.9	83.5	14.1	84.2	12.4	84.9	10.1	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	49.5	52.8	58.8	66.1	73.4	79.6	82.2	81.3	76.8	67.9	58.2	52.6
	DBStd	13.46	9.79	8.77	8.37	6.83	5.69	4.22	3.01	3.29	4.45	7.06	8.19	9.26
	HDD50	322	133	65	23	1	0	0	0	0	0	0	20	80
	HDD65	1821	486	350	224	69	8	0	0	0	1	54	234	395
	CDD50	6404	116	143	295	484	726	889	997	970	804	554	265	161
	CDD65	2429	4	8	31	103	269	439	532	505	355	143	29	11
	CDH74	21551	18	49	219	723	2135	4108	5529	4949	2733	909	139	40
	CDH80	7852	0	2	29	152	691	1649	2321	1944	879	176	9	0
Wind	WSAvg	7.7	8.0	8.4	8.9	9.0	8.3	7.5	7.1	6.7	7.3	7.0	7.2	7.4
Precipitation	PrecAvg	49.3	3.4	3.2	3.6	3.0	2.9	5.6	6.0	6.6	5.3	4.2	2.6	3.0
	PrecMax	65.5	8.0	10.4	6.2	6.3	6.4	11.9	9.9	10.7	11.5	11.3	7.9	8.8
	PrecMin	33.6	0.4	1.3	0.4	1.0	0.4	2.8	2.3	3.5	0.4	0.0	0.0	0.6
	PrecStd	7.8	2.0	2.1	1.6	1.5	1.7	2.5	2.0	2.4	2.7	3.1	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	79.2	83.7	87.7	93.6	96.4	96.9	96.4	92.1	87.2	81.2	78.3
		MCWB	65.6	65.6	65.8	69.4	73.5	77.5	78.4	79.4	75.9	74.4	70.4	68.1
	2%	DB	72.3	75.0	79.4	84.0	89.7	93.5	94.3	93.1	89.3	84.2	77.8	74.5
		MCWB	63.5	63.2	64.6	68.0	72.6	76.7	78.5	78.4	75.5	72.5	67.5	66.1
	5%	DB	68.6	71.0	76.2	81.2	86.5	90.8	92.1	90.9	87.3	81.8	75.0	70.9
		MCWB	61.5	60.5	62.8	67.1	71.9	76.3	78.3	78.1	75.0	71.0	66.1	65.3
	10%	DB	64.9	67.3	72.5	78.4	83.8	88.4	90.0	88.9	85.3	79.3	72.1	67.5
		MCWB	59.0	59.3	61.2	65.6	70.9	75.5	77.9	77.3	74.3	69.6	64.9	62.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.8	68.8	69.6	73.8	77.2	80.4	82.3	82.2	79.3	77.8	73.1	71.0
		MCDB	72.8	75.3	77.3	81.4	86.9	90.8	91.1	90.7	85.7	81.9	76.6	75.0
	2%	WB	65.8	66.2	67.8	71.5	75.4	79.1	80.7	80.5	78.1	75.9	70.8	68.4
		MCDB	70.2	71.7	75.0	79.0	83.6	88.8	89.2	88.2	84.5	80.8	74.7	72.0
	5%	WB	63.3	63.9	65.8	69.8	74.3	78.0	79.8	79.7	77.2	74.3	68.8	66.0
		MCDB	67.5	68.6	72.1	76.9	82.1	86.8	88.1	87.2	83.5	79.1	73.1	70.1
	10%	WB	60.3	61.0	64.0	68.3	73.1	77.1	78.8	78.6	76.3	72.4	66.2	63.3
		MCDB	64.0	65.7	70.0	75.1	80.4	85.1	86.7	85.8	82.1	76.9	70.6	67.2
Mean Daily Temperature Range	5% DB	MDBR	20.1	20.8	21.1	20.9	18.9	16.7	16.1	15.4	15.7	18.7	20.9	19.6
		MCDBR	22.3	23.6	23.8	22.1	20.8	19.2	18.0	17.3	17.4	19.0	21.6	20.5
		MCWBR	15.0	13.8	11.6	9.6	7.8	7.3	6.8	6.5	6.5	9.0	12.4	13.5
	5% WB	MCDBR	19.5	20.4	20.0	18.8	17.8	17.2	16.6	16.0	14.6	14.9	17.5	18.7
		MCWBR	15.1	14.1	11.7	9.7	7.5	7.2	7.0	6.7	6.4	8.4	11.9	14.1
Clear-Sky Solar Irradiance	taub	0.326	0.334	0.365	0.408	0.468	0.499	0.535	0.515	0.436	0.390	0.348	0.331	
	taud	2.535	2.509	2.441	2.303	2.184	2.146	2.058	2.112	2.329	2.434	2.508	2.549	
	Ebn at Noon	284	292	290	280	263	254	244	247	264	269	272	275	
	Edh at Noon	27	30	35	41	47	49	53	49	38	32	27	25	
All-Sky Solar Radiation	RadAvg	903	1113	1472	1881	2030	2060	1978	1785	1519	1307	1005	804	
	RadStd	52	114	132	127	138	143	121	110	128	141	71	65	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+212
(47)	Regional (43 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+371	+259

Nomenclature: See separate page